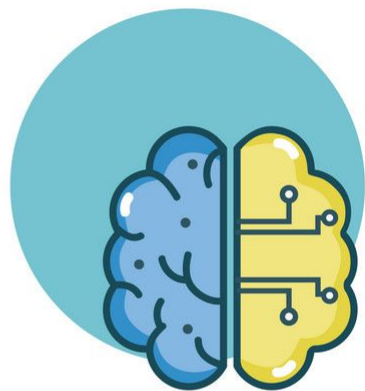


INTRODUCTION TO MACHINE LEARNING

THIS COURSE



Elisa Ricci



THIS COURSE

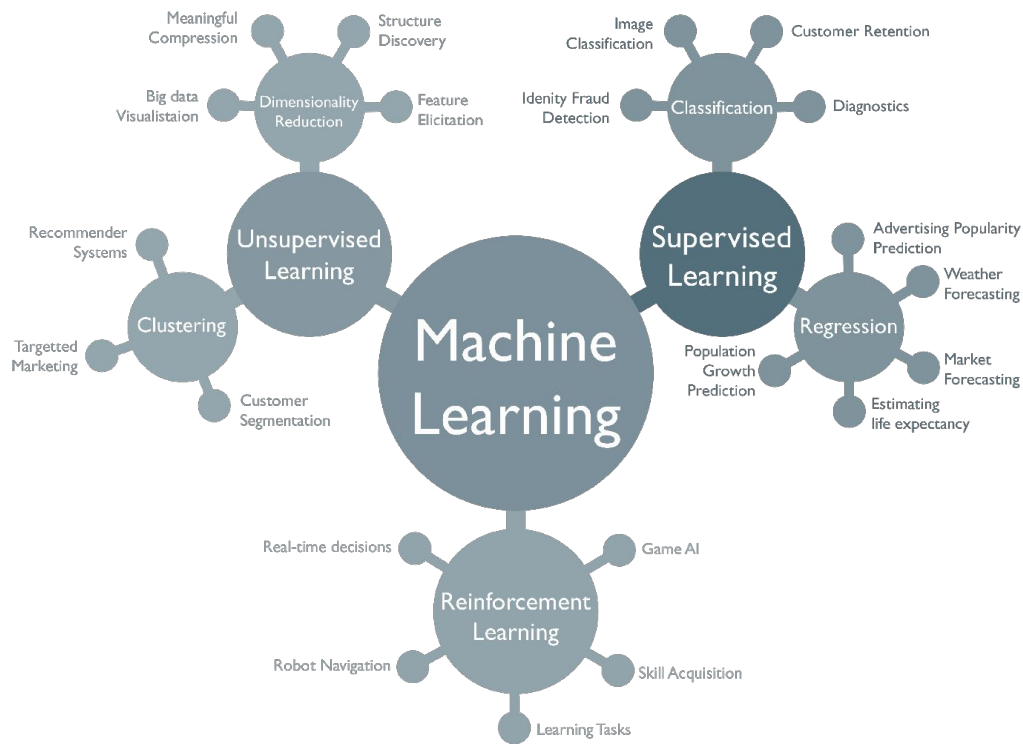
- This course provides a **broad introduction** to machine learning and statistical pattern recognition.
- The course will cover **different topics** ranging from supervised learning (parametric and non-parametric algorithms, kernel methods, feedforward neural networks), unsupervised learning (clustering, dimensionality reduction, autoencoders, deep generative models) and reinforcement learning.
- During the course some **applications** of machine learning will be presented in the field of computer vision, robotics and natural language processing.
- Theoretical discussion will be complemented with **lab in Python** using open-source libraries.

GOALS OF THE COURSE: LEARN ABOUT...

- Different machine learning problems and algorithms
- Common techniques and tools used
 - Theoretical understanding
 - Practical implementation
- Proper experimentation and evaluation
 - Programming tools

CONTENTS

- Introduction to Machine Learning
- Supervised Learning
 - Nearest Neighbors
 - Decision Trees and Random Forests
 - Kernel Methods
 - Ensemble Methods
 - Deep Neural Networks (Feedforward, convolutional and recurrent networks)
- Unsupervised Learning
 - Clustering
 - Dimensionality reduction
 - Deep Generative Models
- Reinforcement Learning



COURSE EXPECTATIONS

- Interesting topics
- A bit of programming
- ... but machine learning involves algebra, optimization

INSTRUCTORS

Elisa Ricci

- Contacts:

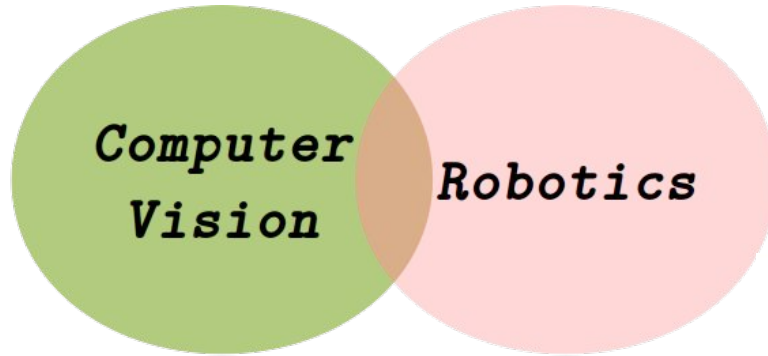
- *mail:* e.ricci@unitn.it
- *phone:* + 39 046128 3778
- *web:* <http://eliricci.eu/>
- *room:* Povo 2, Room 111

- Meeting Times: online (send me an email before!)
- Use the [INTRO2ML] tag in the subject of the email



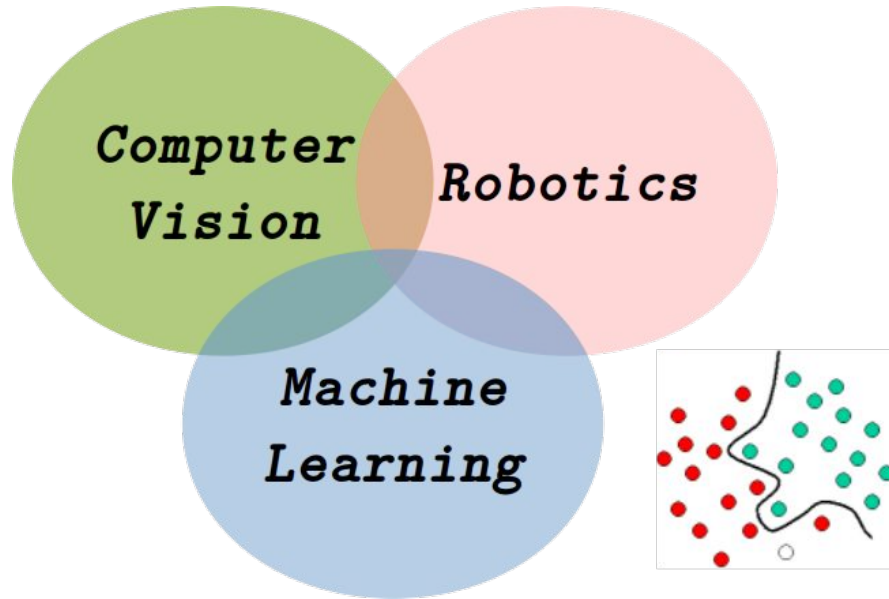
TA: Francesco Gentile, francesco.gentile@unitn.it

ELISA'S BACKGROUND

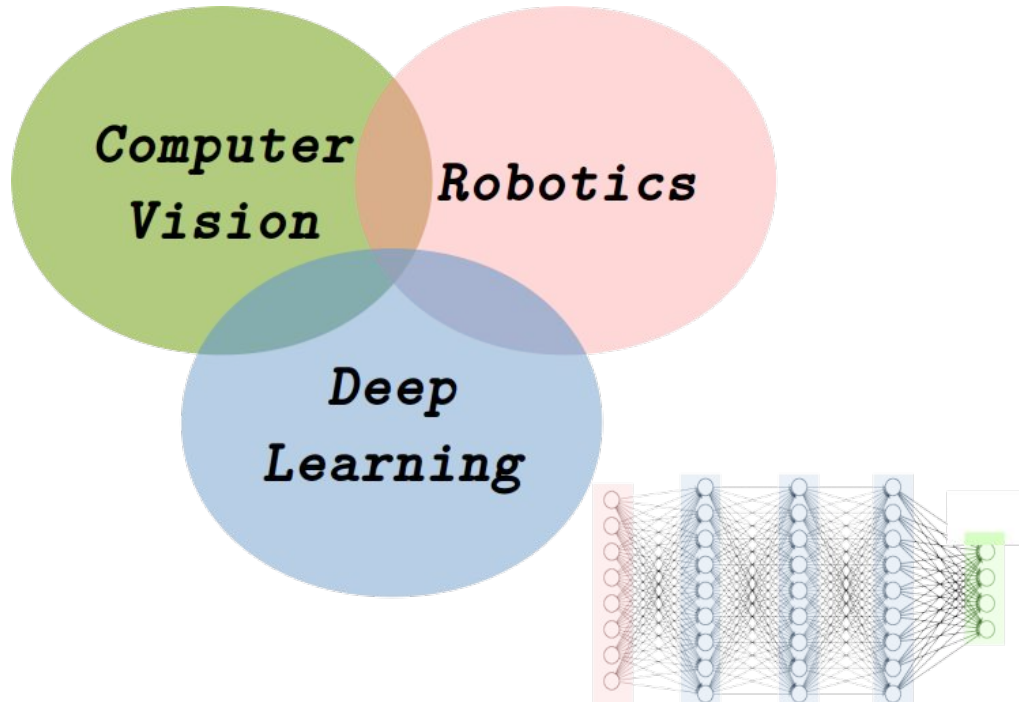


Social Robots

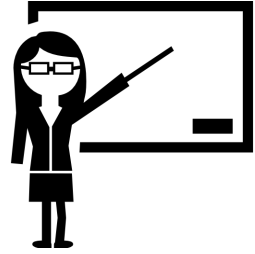
ELISA'S BACKGROUND



ELISA'S BACKGROUND



TEACHING MODALITIES



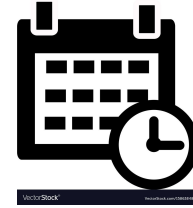
- **Slides** will be used.
- Slides will be made available to students beforehand (esse3).
- **Calendar** available on moodle.
- **Lab exercises** will be carried out in Python with open-source libraries.
- A **guest speaker**, either from industry or from a university, will be invited to give a talk at the end of the course.

EXAMS



- Students will be assessed with an **oral exam**.
- The oral exam will consist of **three questions** about theory.
- Mark is given on a 30-point scale.
- Online exam: <https://dol.unitn.it/esami-distanza>

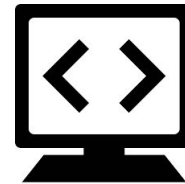
SCHEDULE



- Period:
 - Wednesday 10:30 - 12:30 Room B109
 - Thursday 08:30 - 10:30 Room A101 (discuss to extend one hour)

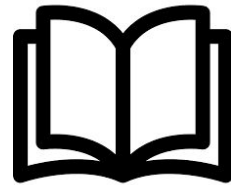
- Variations on schedule will be notified on *Didattica Online* (see also [here](#))

LABS

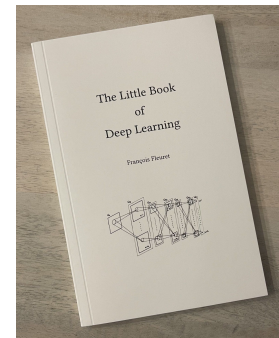
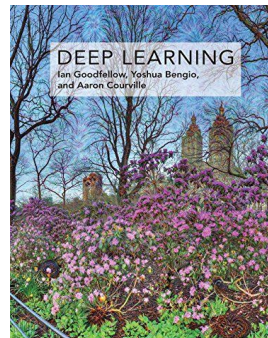
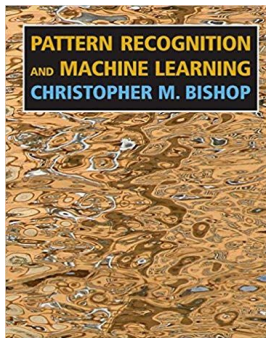


- **Prerequisites:** Basic computer science algorithms, linear algebra, basic optimization, (some python programming)
- **Labs** starting in few weeks!

BOOKS



- C. Bishop, Pattern Recognition and Machine Learning, Springer, 2006
- I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press, 2016.
- S. Prince, Understanding Deep Learning, 2023
- F. Fleuret, The Little Book of Deep Learning, 2023
(<https://fleuret.org/francois/lbdl.html>)
- Lots of online blogs and videos...
- **No generative AI** (you will understand why!)



WHAT DO YOU EXPECT FROM THE COURSE?



QUESTIONS?

